

LLM-DetectAIve: a Tool for Fine-Grained Machine-Generated Text Detection

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Abstract

The ease of access to large language models (LLMs) has enabled a widespread of machine-generated texts, and now it is often hard to tell whether a piece of text was human-written or machine-generated. This raises concerns about potential misuse, particularly within educational and academic domains. Thus, it is important to develop practical systems that can automate the process. Here, we present one such system, **LLM-DetectAIve**, designed for fine-grained detection. Unlike most previous work on machine-generated text detection, which focused on binary classification, LLM-DetectAIve supports four categories: (i) human-written, (ii) machine-generated, (iii) machine-written, then machine-humanized, and (iv) human-written, then machine-polished. Category (iii) aims to detect attempts to obfuscate the fact that a text was machine-generated, while category (iv) looks for cases where the LLM was used to polish a human-written text, which is typically acceptable in academic writing, but not in education. Our experiments show that LLM-DetectAIve can effectively identify the above four categories, which makes it a potentially useful tool in education, academia, and other domains. LLM-DetectAIve is publicly accessible at <https://github.com/mbzuai-nlp/LLM-DetectAIve>.¹ The video describing our system is available at https://youtu.be/E8eT_bE7k8c.

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1 Introduction

The development of advanced large language models (LLMs), such as GPT-4, Claude-3.5, Gemini-1.5, Llama-70b (OpenAI, 2023; Anthropic, 2024; Gemini, 2023; Llama, 2024), improved the prevalence and the coherence of machine-generated content. This trend makes it increasingly difficult to differentiate between texts produced by machines from such written by humans (Macko et al., 2023; Wang et al., 2024b,c). As a result, there have been growing concerns about the authenticity and integrity of textual content (Crothers et al., 2023; Tang et al., 2024).

While many detectors have been developed to address this new challenge (Mitchell et al., 2023; Wang et al., 2024a), they often struggle to keep up with the rapid development of LLMs. Generations produced by new models are hard to detect as they become more coherent and represent out-of-distribution instances, compared to what detecting systems saw during training (Macko et al., 2024; Koike et al., 2024). Moreover, the use of prompting to generate more human-like texts or applying LLMs to refine or change the tone of human writings further complicates detection.

Most prior work on detecting machine-generated text focused on binary detection, i.e., predicting whether the text is generated by a machine or written by a human. This dichotomy leaves no space for mixed categories of human-machine collaboration. However, we argue for the need for additional categories, as machine-polishing of human-written text is acceptable in certain cases (e.g., for academic papers), but not in other (e.g., in education).

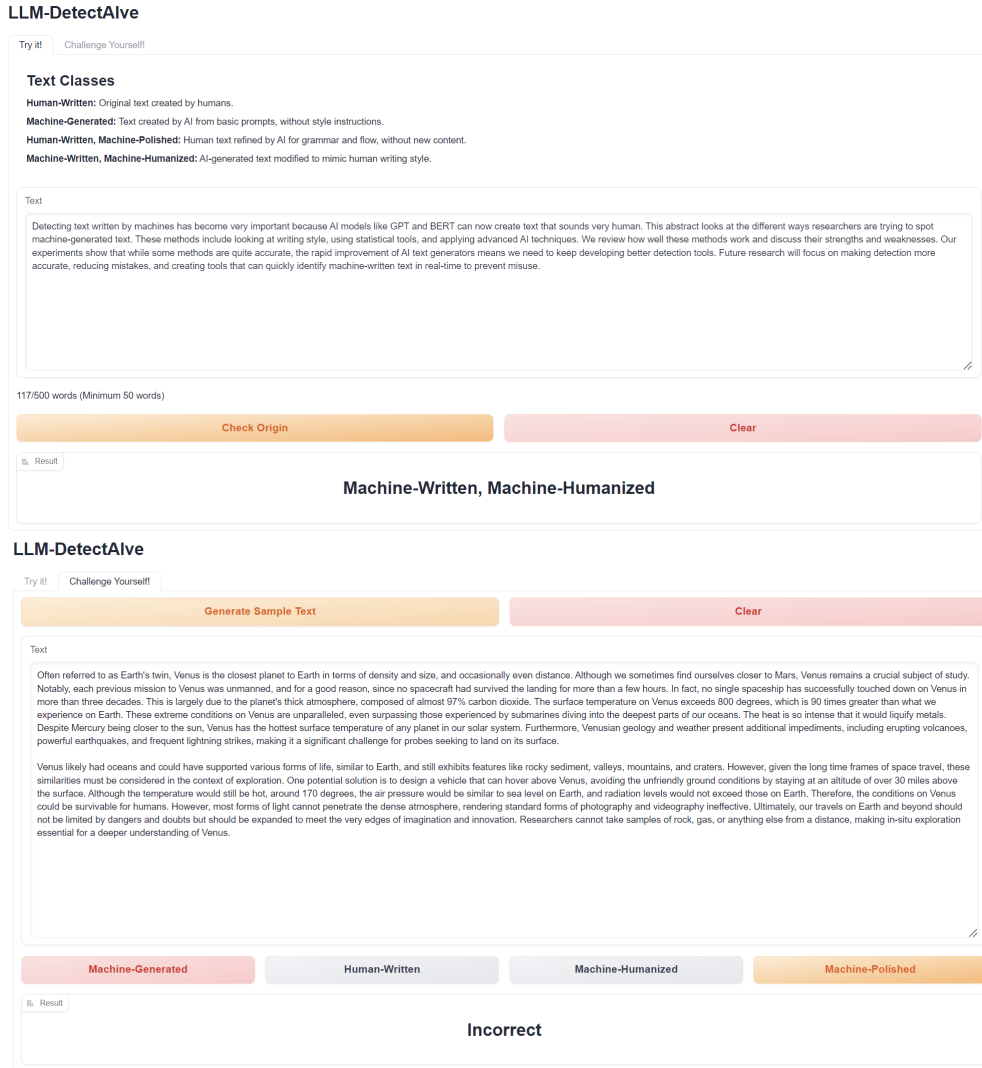


Figure 1: **LLM-DetectAIve interface:** automatic text detection (top) and human detector playground (bottom).

In education, using LLMs to complete entire assignments or even to polish human-written essays is typically prohibited (Susnjak, 2022). Therefore, it is important to perform fine-grained text classification. For example, detecting the use of LLMs in text humanization and refinement becomes critical to ensure the fair assessment of students' genuine knowledge and abilities. Fine-grained human/machine identification is also important for authorship detection in digital forensics. A contemporary work by Zhang et al. (2024) also highlighted the importance of identifying LLM contributions as coauthors in a human-AI collaborative text.

To address this problem, we propose a new formulation of problem, as multi-way classification with the following labels:

- I. **Human-Written:** the ext is created solely by a human author without GenAI assistance.

- II. **Machine-Generated:** the text is entirely produced by a machine based on input prompts without any human intervention.

- III. **Machine-Written Machine-Humanized:** the text is initially generated by a machine and then subtly modified to appear more human-like. This involves automatically tweaking the LLM to make the output appear more human.

- IV. **Human-Written Machine-Polished:** the text is written by a human and then is refined or polished by a machine, e.g., to correct grammar, improve style, and/or optimize readability while trying to preserve the meaning of the original human text.

We further develop **LLM-DetectAIve**, a system that accurately distinguishes between different types of text generation and editing. With this,

Text Class	Generator	OUTFOX	Wikipedia	Wikihow	Reddit ELI5	arXiv abstract	PeerRead
M4GT-Bench							
I	Human	14,043	14,333	15,999	16,000	15,998	2,847
II	davinci-003	3,000	3,000	3,000	3,000	3,000	2,340
	gpt-3.5-turbo	3,000	2,995	3,000	3,000	3,000	2,340
	cohere	3,000	2,336	3,000	3,000	3,000	2,342
	dolly-v2	3,000	2,702	3,000	3,000	3,000	2,344
	BLOOMz	3,000	2,999	3,000	2,999	3,000	2,334
	gpt4	3,000	3,000	3,000	3,000	3,000	2,344
New Generations							
II + III + IV	gpt-4o	8,966	8,995	9,000	9,000	9,000	7,527
	gemma-7b	8,280	8,985	9,000	9,000	9,000	0
	llama3-8b	8,271	8,985	9,000	9,000	9,000	0
	llama3-70b	8,577	8,985	9,000	9,000	9,000	0
	mixtral-8x7b	17,001	8,985	9,000	9,000	9,000	0
	gemma2-9b	0	8,985	9,000	9,000	9,000	0
III	gemini1.5	0	1,652	1,601	904	0	0
	mistral-7b	0	2,993	3,000	0	0	2,344
IV	gemini1.5	0	1,652	1,601	904	2,994	586
	mistral-7b	0	2,993	3,000	0	0	2,344

Table 1: **Statistics about our datasets** across LLMs over the four classes: I. Human-Written, II. Machine-Generated, III. Machine-Written Machine-Humanized and IV. Human-Written Machine-Polished. For row II + III + IV, the data is approximately uniformly distributed across the three classes.

we aim to uphold academic integrity and ensure a fair evaluation process for both students and researchers.

Our contributions are as follows:

- We reformulate the task as fine-grained multi-way classification.
- We collect a dataset for this reformulation using generations from a variety of LLMs.
- We build, evaluate, and compare several machine-generated text detectors on our new fine-grained dataset.
- We develop a Web-based demo that (i) allows users to input text and to obtain fine-grained classification prediction, and (ii) offers a playground for users to test their ability to detect texts with varying degrees of LLM involvement, according to the above 4-way fine-grained schema.

2 Dataset

To collect the dataset for our multi-way fine-grained detector, we first gathered datasets that were curated for binary machine-generated text detection from previous work, and then we extended the data into our four labels by introducing new corresponding generations. Sections 2.2 and 2.3 discuss the prompts we used for generation and data cleaning, respectively.

2.1 Data Overview

We build the new dataset by extending the M4GT-Bench (Wang et al., 2024b), which is an benchmark dataset for evaluating machine-generation text detectors that encompasses multiple generators and domains, including arXiv, Wikihow, Wikipedia, Reddit, student essays (OUTFOX), and peer reviews (PeerRead). From these sources, we sampled a subset comprising 79,220 human-written texts and 103,075 machine-generated texts.

Next, we expanded this dataset by (i) collecting additional machine-generated texts produced by new LLMs (e.g., GPT-4o), (ii) generating machine-written then machine-humanized texts, and (iii) polishing human-written texts using various LLMs. This resulted in 91,358 fully-MGTs, 103,852 machine-written then machine-humanized texts, and 107,900 human-written then machine-polished texts. Table 1 gives detailed statistics about the dataset.

For data generation, we used a variety of LLMs, including Llama3-8b, Llama3-70b (Llama, 2024), Mixtral 8x7b (Jiang et al., 2024), Gemma-7b, Gemma2-9b (Team, 2024), GPT-4o (OpenAI, 2023), Gemini-1.5-pro (Gemini, 2023), and Mistral-7b (Jiang et al., 2023). By incorporating a diverse array of LLMs and domains, we aim to enhance the detection accuracy within actual domains and generators, as well as improve generalization.